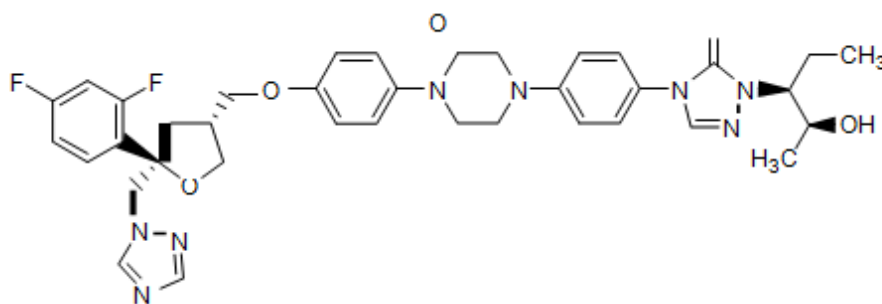


NOXAFIL[®] 40 mg/ml ORAL SUSPENSION Product Information**NAME OF THE MEDICINE**

Posaconazole

Posaconazole is a broad spectrum triazole antifungal compound with a molecular formula of $C_{37}H_{42}F_2N_8O_4$ yielding a molecular weight of 700.8.

The chemical structure, which possesses four chiral centres, two R and two S, and chemical name are illustrated below:



CAS INDEX NAME: D-threo-Pentitol, 2,5-anhydro-1,3,4-trideoxy-2-C-(2,4-difluorophenyl)-4-[[4-[4-[4-[1-[(1S,2S)-1-ethyl-2-hydroxypropyl]-1,5-dihydro-5-oxo-4H-1,2,4-triazol-4-yl]phenyl]-1-piperazinyl]phenoxy]methyl]-1-(1H-1,2,4-triazol-1-yl) CAS RN 171228-49-2.

IUPAC NAME: 4-4-[4-(4-{(3R, 5R)-5-(2,4-difluorophenyl)-5-(1H-1,2,4-triazol-1-yl)methyl}tetrahydro-3-furanyl]methoxyphenyl)piperazino]phenyl-1-[(1S,2S)-1-ethyl-2-hydroxypropyl]-4,5-dihydro-1H-1,2,4-triazol-5-one.

DESCRIPTION

Posaconazole is a white to off-white crystalline powder. It has a melting range of 164° – 165° C and is insoluble in water.

NOXAFIL ORAL SUSPENSION is a white, cherry flavoured immediate-release oral suspension containing 40 mg of posaconazole per mL and the following inactive ingredients: polysorbate 80, simethicone, sodium benzoate, sodium citrate dihydrate, citric acid monohydrate, glycerol, xanthan gum, liquid glucose, titanium dioxide, artificial cherry flavouring, and purified water.

PHARMACOLOGY Pharmacodynamic properties

Pharmacotherapeutic Group

Antiinfective for Systemic Use, Triazole derivative, J02AC04

Mechanism of action:

Posaconazole is a triazole antifungal agent. It is an inhibitor of the enzyme lanosterol 14 α -demethylase, which catalyses an essential step in ergosterol biosynthesis. Ergosterol depletion, coupled with the accumulation of methylated sterol precursors, is thought to impair membrane integrity and the function of some membrane-associated proteins. This results in the inhibition of cell growth and/or cell death.

Consequently, posaconazole exhibits broad-spectrum antifungal activity against a variety of yeasts and moulds including species of *Candida* (including *C. albicans* isolates resistant to fluconazole, voriconazole and itraconazole, *C. krusei* and *C. glabrata* which are inherently less susceptible to fluconazole, and *C. lusitaniae* which is inherently less susceptible to amphotericin B), *Aspergillus* (including isolates resistant to fluconazole, voriconazole, itraconazole and amphotericin B) and organisms not previously regarded as being susceptible to azoles such as the zygomycetes (e.g. species of *Absidia*, *Mucor*, *Rhizopus* and *Rhizomucor*).

In vitro posaconazole exhibited fungicidal activity against species of *Aspergillus*, dimorphic fungi (*Blastomyces dermatitidis*, *Histoplasma capsulatum*, *Penicillium marneffe*, and *Coccidioides immitis*) and some species of *Candida*. In animal infection models posaconazole was active against a wide variety of fungal infections caused by moulds or yeasts. However, there was no consistent correlation between minimum inhibitory concentration and efficacy.

Microbiology:

Posaconazole has been shown *in vitro* and in clinical infections to be active against the following microorganisms: (See Indications): *Aspergillus* species (*Aspergillus fumigatus*, *A. flavus*, *A. terreus*, *A. nidulans*, *A. niger*, *A. ustus*, *A. ochraceus*), *Candida* species (*Candida albicans*, *C. glabrata*, *C. krusei*, *C. parapsilosis*), *Cryptococcus neoformans*, *Coccidioides immitis*, *Fonsecaea pedrosoi*, *Histoplasma capsulatum*, *Pseudallescheria boydii* and species of *Alternaria*, *Exophiala*, *Fusarium*, *Ramichloridium*, *Rhizomucor*, *Mucor*, and *Rhizopus*. While posaconazole has been used in a clinical setting against these microorganisms, sufficient evidence for efficacy has not been collected for all the listed microorganisms (see Clinical Trials).

Posaconazole also exhibits *in vitro* activity against the following yeasts and moulds: *Candida dubliniensis*, *C. famata*, *C. guilliermondii*, *C. lusitaniae*, *C. kefir*, *C. rugosa*, *C. tropicalis*, *C. zeylanoides*, *C. inconspicua*, *C. lipolytica*, *C. norvegensis*, *C. pseudotropicalis*, *Cryptococcus laurentii*, *Kluyveromyces marxianus*, *Saccharomyces cerevisiae*, *Yarrowia lipolytica*, species of *Pichia*, and *Trichosporon*, *Aspergillus sydowii*, *Bjerkandera adusta*, *Blastomyces dermatitidis*, *Epidermophyton floccosum*, *Paracoccidioides brasiliensis*, *Scedosporium apiospermum*, *Sporothrix schenckii*, *Wangiella dermatitidis* and species of *Absidia*, *Apophysomyces*, *Bipolaris*, *Curvularia*, *Microsporum*, *Paecilomyces*, *Penicillium*, and *Trichophyton*. However, the safety and effectiveness of posaconazole in treating clinical infections due to these microorganisms have not been established in clinical trials.

The following *in vitro* data are available, but their clinical significance is unknown. In a surveillance study of > 3,000 clinical mould isolates from 2010-2018, 90 % of non-*Aspergillus* fungi exhibited the following *in vitro* minimum inhibitory concentration (MIC): *Mucorales* spp (n=81) of 2 mg/L; *Scedosporium apiospermum*/*S. boydii* (n=65) of 2 mg/L; *Exophiala dermatitidis* (n=15) of 0.5 mg/L, and *Purpureocillium lilacinum* (n=21) of 1 mg/L.

NOXAFIL exhibits broad-spectrum antifungal activity against some yeasts and moulds not generally responsive to azoles, or resistant to other azoles:

- species of *Candida* (including *C. albicans* isolates resistant to fluconazole, voriconazole and itraconazole,
- *C. krusei* and *C. glabrata* which are inherently less susceptible to fluconazole,
- *C. lusitaniae* which is inherently less susceptible to amphotericin B),

- *Aspergillus* (including isolates resistant to fluconazole, voriconazole, itraconazole and amphotericin B)
- organisms not previously regarded as being susceptible to azoles such as the zygomycetes (e.g. species of *Absidia*, *Mucor*, *Rhizopus* and *Rhizomucor*).
- *In vitro* **NOXAFIL** exhibited fungicidal activity against species of:
 - *Aspergillus*,
 - dimorphic fungi (*Blastomyces dermatitidis*, *Histoplasma capsulatum*,
 - *Penicillium marneffe*,
 - *Coccidioides immitis*)
 - some species of *Candida*.

In animal infection models **NOXAFIL** was active against a wide variety of fungal infections caused by moulds or yeasts. However, there was no consistent correlation between minimum inhibitory concentration and efficacy.

Specimens for fungal culture and other relevant laboratory studies (including histopathology) should be obtained prior to therapy to isolate and identify causative organism(s). Therapy may be instituted before the results of the cultures and other laboratory studies are known. However, once these results become available, antifungal therapy should be adjusted accordingly.

Drug Resistance:

C. albicans strains resistant to posaconazole could not be generated in the laboratory; spontaneous laboratory *Aspergillus fumigatus* mutants exhibiting a decrease in susceptibility to posaconazole arose at a frequency of 1×10^{-8} to 1×10^{-9} . Clinical isolates of *Candida albicans* and *Aspergillus fumigatus* exhibiting significant decreases in posaconazole susceptibility are rare. In those rare instances where decreased susceptibility was noted, there was no clear correlation between decreased susceptibility and clinical failure. Clinical success has been observed in patients infected with organisms resistant to other azoles; consistent with these observations posaconazole was active *in vitro* against many *Aspergillus* and *Candida* strains that developed resistance to other azoles and/or amphotericin B. Breakpoints for posaconazole have not been established for any fungi.

Antifungal drug combinations:

When combinations of posaconazole with either amphotericin B or caspofungin were tested *in vitro* and *in vivo* there was little or no antagonism and in some instances there

was an additive effect. Clinical studies of posaconazole in combination with antifungal drugs including amphotericin B-based drugs and caspofungin have not been conducted.

Pharmacokinetic Properties

Absorption:

Posaconazole is absorbed with a median T_{max} of 3 hours (patients) and ~ 5 hours (healthy volunteers). Intersubject variability in mean AUC and C_{max} was high in healthy volunteers and patients despite the controlled conditions in pharmacokinetic studies.

The pharmacokinetics of posaconazole are linear following single and multiple dose administration of up to 800 mg. No further increases in exposure are observed above a total daily dose of 800 mg in patients and healthy volunteers. There is no effect of altered pH on the absorption of posaconazole (See Drug Interactions).

Dividing the total posaconazole daily dose (800 mg) as 400 mg twice a day results in a 184 % higher exposure relative to once-a-day administration in patients. Exposure further increased when posaconazole was given as 200 mg four times daily.

Effect of food on oral absorption healthy volunteers:

The AUC of posaconazole is about 2.6 times greater when administered with a nonfat meal or nutritional supplement (14 gm fat) and 4 times greater when administered with a high-fat meal (~ 50 gm fat) relative to the fasted state.

Posaconazole should be administered with food or a nutritional supplement (See Dosage and Administration).

Distribution:

Posaconazole has a large apparent volume of distribution (1774 L) suggesting extensive penetration into the peripheral tissues. Posaconazole is highly protein bound (> 98.0 %), predominantly to serum albumin.

Metabolism:

Posaconazole does not have any major circulating metabolites and its concentrations are unlikely to be altered by inhibitors of CYP450 enzymes. Of the circulating metabolites, the majority are glucuronide conjugates of posaconazole with only minor amounts of oxidative (CYP450 mediated) metabolites observed. The excreted metabolites in urine and faeces account for approximately 17 % of the administered radio-labelled dose.

Excretion:

Posaconazole is slowly eliminated with a mean half-life ($t_{1/2}$) of 35 hours (range 20 to 66 hours) and a total body clearance (Cl/F) of 32 L/hr. Posaconazole is predominantly excreted in the faeces (77 % of the radio-labelled dose) with the major component eliminated as parent drug (66 % of the radio-labelled dose). Renal clearance is a minor elimination pathway, with 14 % of the radio-labelled dose excreted in urine (< 0.2 % of the radio-labelled dose is parent drug). Steady-state is attained following 7 to 10 days of multiple-dose administration.

Summary of the mean pharmacokinetic parameters in patients:

The general pharmacokinetic findings across the clinical program in both healthy volunteers and patients were consistent, in that posaconazole was slowly absorbed and slowly eliminated with an extensive volume of distribution. In addition, the phenomenon of dose-limited absorption of posaconazole at 800 mg/day was observed both in healthy volunteers and patients. The mean pharmacokinetic parameters in patients and healthy volunteers following administration of posaconazole 400 mg twice a day for 7 days are displayed in Table 1.

TABLE 1 - Pharmacokinetics of posaconazole in patients and healthy volunteers

Population	Dose	Mean (%CV)		
		C _{max} (ng/mL)	T _{max} ^a (hr)	AUC(T) (ng·hr/mL)
Healthy Volunteers	400 mg twice a day (n=174)	2850 (36)	5 (0-12)	29453 (37)
Patients	400 mg twice a day (n=24)	851 (82)	3 (0-12.5)	8619 (86)

a: Median (range)

The exposure to posaconazole following administration of 400 mg twice a day was~ 3 times higher in healthy volunteers than in patients, without additional safety findings at the higher concentrations (Table 1).

Pharmacokinetics in Special Populations:Children (< 18 years)

There is limited paediatric experience for posaconazole tablets.

Following administration of 800 mg per day of posaconazole as a divided dose for treatment of invasive fungal infections, mean trough plasma concentrations from 12 patients 8 – 17 years of age (776 ng/ml) were similar to concentrations from 194 patients 18 – 64 years of age (817 ng/ml). No pharmacokinetic data are available from paediatric patients less than 8 years of age. Similarly, in the prophylaxis studies, the mean steady-state posaconazole average concentration (C_{av}) was comparable among ten adolescents (13-17 years of age) to C_{av} achieved in adults (≥ 18 years of age).

In a study of 136 neutropenic paediatric patients 11 months – 17 years treated with posaconazole oral suspension, at doses up to 18 mg/kg/day divided TID, approximately 50% met the prespecified target (Day 7 C_{avg} between 500 ng/mL-2500 ng/mL).

In general, exposures tended to be higher in the older patients (7 to <18 years) than in younger patients (2 to <7 years).

Gender

The pharmacokinetics of posaconazole are comparable in men and women. No adjustment in the dosage of NOXAFIL is necessary based on gender.

Elderly

Results from a multiple dose study in healthy volunteers ($n = 48$) indicated that at steady state, there was an increase in C_{max} (26 %) and AUC (29 %) observed in elderly subjects (24 subjects ≥ 65 years of age) relative to younger subjects (24 subjects 18 -45 years of age). A similar trend was observed in the clinical program based on a small proportion of elderly subjects ≥ 65 years of age ($n=25$ vs. 194 patients 18 –64 years of age). However, in a population pharmacokinetic analysis (Study 1899), age did not influence the pharmacokinetics of posaconazole. The safety profile of posaconazole between the young and elderly patients was similar. Therefore no dose adjustment is required for age.

Race

Results from a multiple dose study in healthy volunteers ($n = 56$) indicated that there was only a slight decrease (16 %) in the AUC and C_{max} of posaconazole in Black subjects relative to Caucasian subjects, therefore, no dose adjustment for race is required.

Renal insufficiency

Following single-dose administration, there was no effect of mild and moderate renal

insufficiency ($n=18$, $Cl_{cr} \geq 20 \text{ mL/min/1.73 m}^2$) on posaconazole pharmacokinetics, therefore, no dose adjustment is required. In subjects with severe renal insufficiency ($n=6$, $Cl_{cr} < 20 \text{ mL/min/1.73 m}^2$), the exposure of posaconazole was highly variable (96 % CV) compared to the exposure in the other renal groups (40 % CV). However, as posaconazole is not significantly renally eliminated, an effect of severe renal insufficiency on the pharmacokinetics of posaconazole is not expected and no dose adjustment is recommended. Posaconazole is not removed by haemodialysis.

Hepatic insufficiency

In a small number of subjects ($n=12$) studied with hepatic insufficiency (Child-Pugh class A, B or C), C_{max} values generally decreased with the severity of hepatic dysfunction (545, 414 and 347 ng/mL for the mild, moderate, and severe groups, respectively), even though the C_{max} values (mean 508 ng/mL) for the normal subjects were consistent with previous trials in healthy volunteers. In addition, an increase in half-life was also associated with a decrease in hepatic function (26.6, 35.3, and 46.1 hours for the mild, moderate, and severe groups, respectively), as all groups had longer half-life values than subjects with normal hepatic function (22.1 hours). Due to the limited pharmacokinetic data in patients with hepatic insufficiency; no recommendation for dose adjustment can be made.

Electrocardiogram evaluation:

Multiple, time-matched ECGs collected over a 12 hour period were recorded at baseline and steady-state from 173 healthy male and female volunteers (18 to 85 years of age) administered posaconazole 400 mg BID with a high-fat meal. In this pooled analysis, the mean QT_c (Fridericia) interval change was -5 msec following administration of the recommended clinical dose. A decrease in the QT_c (F) interval (- 3 msec) was also observed in a small number of subjects ($n=16$) administered placebo. No subject administered posaconazole had a QT_c (F) interval of ≥ 500 msec or an increase ≥ 60 msec in their QT_c (F) interval from baseline.

CLINICAL TRIALS

Invasive Aspergillosis:

Efficacy in patients with refractory disease or intolerance to prior therapy: The efficacy and survival benefit of oral posaconazole for the treatment of invasive aspergillosis in patients with disease refractory to amphotericin B (including liposomal formulations),

itraconazole or, in a small number of cases, voriconazole or echinocandins, and/or with intolerance to amphotericin B (including liposomal formulations) or itraconazole was demonstrated in 107 patients enrolled in a salvage therapy trial. Patients were administered posaconazole 800 mg/day in divided doses for up to 585 days. The median duration of posaconazole therapy was 56 days (1 – 585 days).

The majority of patients were severely immunocompromised with underlying conditions such as haematologic malignancies, including bone marrow transplantation; solid organ transplantation; solid tumours and/or AIDS. An independent expert panel reviewed all patient data, including diagnosis of invasive aspergillosis, refractoriness and intolerance to previous therapy, and clinical outcome in a parallel and blinded fashion with an external control group of 86 patients treated with standard salvage therapy (e.g. amphotericin B including liposomal formulations, and/or itraconazole) mostly at the same time and at the same sites as the patients enrolled in the posaconazole trial.

A success was defined as either complete resolution (complete response) or a clinically meaningful improvement (partial response) of all signs, symptoms and radiographic findings attributable to the fungal infection. Stable, non-progressive disease and failure were considered to be a non-success. Most of the cases of aspergillosis were considered to be refractory in both the posaconazole group (88 %) and in the external control group (79 %) while the remaining patients were intolerant to prior antifungal therapy (12 %, posaconazole; 21 % external control group).

As shown in Table 2, a successful global response at end of treatment was seen in 42 % of posaconazole-treated patients compared to 26 % of the external group (P=0.006).

TABLE 2. Overall Efficacy of Posaconazole at the End of Treatment* for Invasive Aspergillosis in Comparison to an External Control Group

	Posaconazole	External Control Group
Overall Response	45/107 (42 %)	22/86 (26 %)
	Adjusted Odds Ratio ** 4.06 (95 % CI: 1.50, 11.04) P=0.006	
	P=0.006	
	Unadjusted Odds ratio 2.11 (95 % CI: 1.14, 3.92) P=0.018	
Survival at day 365	(38 %)	(22 %)
All mycologically confirmed <i>Aspergillus</i> spp.***	34/76 (45 %)	19/74 (26 %)
<i>A. fumigatus</i>	12/29 (41 %)	12/34 (35 %)
<i>A. flavus</i>	10/19 (53 %)	3/16 (19 %)
<i>A. terreus</i>	4/14 (29 %)	2/13 (15 %)

<i>A. niger</i>	3/5	(60 %)	2/7	(29 %)
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* end of all study drug therapy plus 7 days within 372 days of the start of salvage therapy

** adjusted odds ratio was obtained using a logistic regression model adjusting for major covariates

*** includes other less common species or species unknown

Other Serious Fungal Pathogens

Posaconazole has been shown to be effective against the following additional pathogens when other therapy had been ineffective or when the patient had developed intolerance of the prior therapy:

Zygomycosis: Successful responses to posaconazole therapy were noted in 7/13 (54%) of patients with zygomycete infections. Sites of infection included the sinuses, lung, and skin. Organisms included *Rhizopus*, *Mucor* and *Rhizomucor*. Most of the patients had underlying haematological malignancies, half of which required a bone marrow transplant. Half of the patients were enrolled with intolerance to previous therapy and the other half as a result of disease that was refractory to prior therapy. Three patients were noted to have disseminated disease, one of which had a successful outcome after failing amphotericin B therapy.

Fusarium spp.: Successful responses to posaconazole therapy were seen in 11 of 24 (46%) of patients with fusariosis. Four of the responders had disseminated disease and one patient had disease localized to the eye; the remainder had a variety of sites of infection. Seven of 24 patients had profound neutropenia at baseline. In addition, 3/5 patients with infection due to *F. solani* which is typically resistant to most antifungal agents, were successfully treated.

Chromoblastomycosis/Mycetoma: Successful responses to posaconazole therapy were seen in 9 of 11 (82%) of patients with chromoblastomycosis or mycetoma.

Five of these patients had chromoblastomycosis due to *Fonsecaea pedrosoi* and 4 had mycetoma, mostly due to *Madurella* species.

Coccidioidomycosis: The efficacy of posaconazole in the primary treatment of non-meningeal coccidioidomycosis was demonstrated in 15 clinically evaluable patients enrolled in an open-label, non-comparative trial to receive posaconazole 400 mg daily for 6 months. Most patients were otherwise healthy and had infections at a variety of sites. A satisfactory response (defined as an improvement of at least 50 % of the Cocci score as defined by the BAMSG Coccidioidomycosis trial group) was seen in 12 of 15 patients (80 %) after an average of 4 months of posaconazole treatment. In a separate open-label, non-comparative trial, the safety and efficacy of posaconazole 400 mg twice a day was

assessed in 16 patients with coccidioidomycosis infection refractory to standard treatment.

Most had been treated with amphotericin B (including lipid formulations) and/or itraconazole or fluconazole for months to years prior to posaconazole treatment. At the end of treatment with posaconazole, a satisfactory response (complete or partial resolution of signs and symptoms present at baseline) as determined by an independent panel was achieved for 11/16 (69 %) of patients. One patient with CNS disease that had failed fluconazole therapy had a successful outcome following 12 months of posaconazole therapy.

Treatment of Azole-Susceptible Oropharyngeal Candidiasis (OPC) in HIV- infected patients

A randomised, double-blind, controlled study was completed in HIV-infected patients with azole-susceptible oropharyngeal candidiasis. The primary efficacy variable was the clinical success rate (defined as cure or improvement) after 14 days of treatment. Patients were treated with posaconazole or fluconazole oral suspension (both posaconazole and fluconazole were given as follows: 100 mg twice a day for 1 day followed by 100 mg once a day for 13 days).

The clinical and mycological response rates from the above study are shown in Table 3 below. Posaconazole and fluconazole demonstrated equivalent clinical success rates at Day 14 as well as 4 weeks after the end of treatment. However, posaconazole demonstrated a significantly better mycological response rate than fluconazole 4 weeks after the end of treatment.

TABLE 3. Clinical Success Rates and Mycological Response Rates in Oropharyngeal Candidiasis

Endpoint	Posaconazole	Fluconazole
Clinical Success Rate at Day 14	91.7 % (155/169)	92.5 % (148/160)
Clinical Success Rate 4 Weeks After End of Treatment	68.5 % (98/143)	61.8 % (84/136)
Mycological Response Rate 4 Weeks After End of Treatment*	40.6 % (41/101)	26.4 % (24/91)

*Statistically significant (P=0.0376)

Clinical success rate was defined as the number of cases assessed as having a clinical response (cure or improvement) divided by the total number of cases eligible for analysis. Mycological response rate was defined as mycological success (≤ 20 CFU/ml) divided by the total number of cases eligible for analysis.

Treatment of Oropharyngeal Candidiasis refractory to itraconazole and fluconazole

(rOPC) in HIV-infected patients

The primary efficacy parameter in the short-term treatment study was the clinical success rate (cure or improvement) after 4 weeks of treatment. HIV-infected patients were treated with posaconazole 400 mg twice a day with an option for further treatment during a 3-month maintenance period. A 75 % (132/176) clinical success rate and a 36.5 % (46/126) mycological response rate (≤ 20 CFU/ml) were achieved after 4 weeks of posaconazole treatment. Clinical success rates ranged from 71 % to 100 %, inclusive, for all azole-resistant *Candida* species identified at baseline, including *C. glabrata* and *C. krusei*.

In the long-term treatment study the primary efficacy endpoint was the clinical success rate (cure or improvement) after 3 months of treatment. A total of 100 HIV- infected patients with OPC and/or EC were treated with posaconazole 400 mg twice a day for up to 15 months. Sixty of these patients had been previously treated in Study 330. An 85.6 % (77/90) clinical success rate overall (cure or improvement) was achieved after 3 months of posaconazole treatment; 80.6 % (25/31) for previously untreated subjects.

The mean exposure to posaconazole based on the actual days dosed was 102 days (range: 1-544 days). Sixty-seven percent (67 %, 10/15) of patients treated with posaconazole for at least 12 months had continued clinical success at the last assessment.

Prophylaxis of Invasive Fungal Infections (IFIs) (Studies 316 and 1899):

Two randomised, controlled prophylaxis studies were conducted among patients at high risk for developing invasive fungal infections.

Study 316 was a randomised, double-blind trial of posaconazole oral suspension (200 mg three times a day) versus fluconazole capsules (400 mg once daily) in allogeneic hematopoietic stem cell transplant recipients with graft-versus-host disease (GVHD). The primary efficacy endpoint was the incidence of proven/probable IFIs at 16 weeks post-randomization as determined by an independent, blinded external expert panel. A key secondary endpoint was the incidence of proven/probable IFIs during the on-treatment period (first dose to last dose of study medicinal product + 7 days). The majority (377/600, [63 %]) of patients included had Acute Grade 2 or 3 or chronic extensive (195/600, [32.5 %]) GVHD at study start. The mean duration of therapy was 80 days for posaconazole and 77 days for fluconazole.

Study 1899 was a randomised, evaluator-blinded study of posaconazole oral suspension

(200 mg three times a day) versus fluconazole suspension (400 mg once daily) or itraconazole oral solution (200 mg twice a day) in neutropenic patients who were receiving cytotoxic chemotherapy for acute myelogenous leukemia or myelodysplastic syndromes. The primary efficacy endpoint was the incidence of proven/probable IFIs as determined by an independent, blinded external expert panel during the on-treatment period. A key secondary endpoint was the incidence of proven/probable IFIs at 100 days post-randomization. New diagnosis of acute myelogenous leukemia was the most common underlying condition (435/602, [72 %]). The mean duration of therapy was 29 days for posaconazole and 25 days for fluconazole/itraconazole.

In both prophylaxis studies, aspergillosis was the most common breakthrough infection. See Table 4 and 5 for results from both studies. There were fewer breakthrough *Aspergillus* infections in patients receiving posaconazole prophylaxis when compared to control patients.

Table 4. Results from clinical studies in prophylaxis of Invasive Fungal Infections.

Study	Posaconazole	Control ^a	P-Value
Proportion (%) of patients with proven/probable IFIs			
On-treatment period^b			
1899 ^d	7/304 (2)	25/298 (8)	0.0009
316 ^e	7/291 (2)	22/288 (8)	0.0038
Fixed-time period^c			
1899 ^d	14/304 (5)	33/298 (11)	0.0031
316 ^d	16/301 (5)	27/299 (9)	0.0740

FLU = fluconazole; ITZ = itraconazole; POS = posaconazole.

a: FLU/ITZ (1899); FLU (316).

b: In 1899 this was the period from randomization to last dose of study medicinal product plus 7 days; in 316 it was the period from first dose to last dose of study medicinal product plus 7 days.

c: In 1899, this was the period from randomization to 100 days post-randomization; in 316 it was the period from the baseline day to 111 days post-baseline.

d: All randomized

e: All treated

Table 5. Results from clinical studies in prophylaxis of Invasive Fungal Infections.

Study	Posaconazole	Control ^a
Proportion (%) of Patients with Proven/Probable Aspergillosis		
On-treatment period^b		
1899 ^d	2/304 (1)	20/298 (7)
316 ^e	3/291 (1)	17/288 (6)
Fixed-time period^c		
1899 ^d	4/304 (1)	26/298 (9)
316 ^d	7/301 (2)	21/299 (7)

FLU = fluconazole; ITZ = itraconazole; POS = posaconazole.

a: FLU/ITZ (1899); FLU (316).

b: In 1899 this was the period from randomization to last dose of study medicinal product plus 7 days; in 316 it was the period from first dose to last dose of study medicinal product plus 7 days.

c: In 1899, this was the period from randomization to 100 days post-randomization; in 316 it was the period from the baseline day to 111 days post-baseline.

d: All randomized

e: All treated

In Study 1899, a significant decrease in all cause mortality in favour of posaconazole was observed [POS 49/304 (16 %) vs. FLU/ITZ 67/298 (22 %) p= 0.048]. Based on Kaplan-Meier estimates, the probability of survival up to day 100 after randomization, was significantly higher for posaconazole recipients; this survival benefit was demonstrated when the analysis considered all causes of death (P= 0.0354) as well as IFI-related deaths (P = 0.0209).

In Study 316, overall mortality was similar (POS, 25 %; FLU, 28 %); however, the proportion of IFI-related deaths was significantly lower in the POS group (4/301) compared with the FLU group (12/299; P= 0.0413).

Use in paediatric patients

Sixteen patients 8-17 years of age were treated with 800 mg/day in a study for invasive fungal infections. Based on the available data in 16 of these paediatric patients, the safety

profile appears to be similar to patients ≥ 18 years of age.

Additionally, twelve patients 13-17 years of age received 600 mg/day for prophylaxis of invasive fungal infections (Studies 316 and 1899). The safety profile in these patients < 18 years of age appears similar to the safety profile observed in adults. Based on pharmacokinetic data in 10 of these paediatric patients, the pharmacokinetic profile appears to be similar to patients ≥ 18 years of age.

Safety and efficacy in paediatric patients below the age of 18 years have not been established.

Electrocardiogram evaluation

Multiple, time-matched ECGs collected over a 12 hour period were obtained before and during administration of posaconazole (400 mg twice daily with high fat meals) from 173 healthy male and female volunteers aged 18 to 85 years. No clinically relevant changes in the mean QTc (Fridericia) interval from baseline were observed.

INDICATIONS

NOXAFIL (posaconazole) is indicated for use in the treatment of the following fungal infections in adults:

- Invasive aspergillosis in patients with disease that is refractory to amphotericin B or itraconazole or in patients who are intolerant of these medicinal products;
- Fusariosis, zygomycosis, coccidioidomycosis, chromoblastomycosis, and mycetoma in patients with disease that is refractory to other therapy or patients who are intolerant of other therapy.
- Oropharyngeal candidiasis in patients who are immunocompromised adults, including patients with disease that is refractory to itraconazole and fluconazole.

Refractoriness is defined as progression of infection or failure to improve after a minimum of 7 days of prior therapeutic doses of effective antifungal therapy.

NOXAFIL is also indicated for prophylaxis of invasive fungal infections in the following adult patients:

- Patients receiving remission-induction chemotherapy for acute myelogenous leukaemia (AML) or myelodysplastic syndromes (MDS) expected to result in prolonged neutropenia and who are at high-risk of developing invasive fungal

infections.

- Haematopoietic stem cell transplant (HSCT) recipients who are undergoing high-dose immunosuppressive therapy for graft versus host disease and who are at high risk of developing invasive fungal infections.

CONTRAINDICATIONS

NOXAFIL is contraindicated in patients with known hypersensitivity to posaconazole or to any of the excipients.

Coadministration of posaconazole and ergot alkaloids (ergotamine, dihydroergotamine) is contraindicated as posaconazole may increase the plasma concentration of ergot alkaloids, which may lead to ergotism (See Interactions).

Although not studied *in vitro* or *in vivo*, coadministration of posaconazole and certain drugs metabolized through the CYP3A4 system: terfenadine, astemizole, cisapride, pimozide, and quinidine may result in increased plasma concentrations of those drugs, leading to potentially serious and/or life threatening adverse events, such as QT prolongation and rare occurrences of torsade de pointes (See Interactions).

Coadministration with the HMG-CoA reductase inhibitors that are primarily metabolized through CYP3A4 is contraindicated since increased plasma concentration of these drugs can lead to rhabdomyolysis.

PRECAUTIONS

Hypersensitivity:

There is no information regarding cross-sensitivity between posaconazole and other azole antifungal agents. Caution should be used when prescribing posaconazole to patients with hypersensitivity to other azoles. Subjects with severe or serious reactions to azoles were excluded from key studies of posaconazole.

Hepatic toxicity:

Hepatic reactions (e.g., mild to moderate elevations in ALT, AST, alkaline phosphatase, total bilirubin, and/or clinical hepatitis) have been reported during treatment with posaconazole. Elevated liver function tests were generally reversible on discontinuation of therapy and in some instances these tests normalized without interruption of therapy. Rarely, more severe hepatic reactions with fatal outcomes have been reported.

Posaconazole should be used with caution in patients with severe hepatic impairment. In these patients, the prolonged elimination half-life may lead to increased exposure.

QT prolongation:

Some azoles have been associated with prolongation of the QT_c interval.

Posaconazole must not be administered with medicines that are substrates for CYP3A4 and are known to prolong the QT_c interval (See Contraindications, Interactions, Clinical Trials – Electrocardiogram Evaluation). Noxafil should be administered with caution to patients with pro-arrhythmic conditions such as:

- Congenital or acquired QT_c prolongation
- Cardiomyopathy, especially in the presence of cardiac failure
- Sinus bradycardia
- Existing symptomatic arrhythmias
- Concomitant use with medicinal products known to prolong the QT_c interval

Electrolyte Disturbances: Especially those involving potassium, magnesium or calcium levels, should be monitored and corrected as necessary before and during posaconazole therapy.

Vincristine Toxicity: Concomitant administration of azole antifungals, including posaconazole, with vincristine has been associated with neurotoxicity and other serious adverse reactions, including seizures, peripheral neuropathy, syndrome of inappropriate antidiuretic hormone secretion, and paralytic ileus. Reserve azole antifungals, including posaconazole, for patients receiving a vinca alkaloid, including vincristine, who have no alternative antifungal treatment options (see Interactions with Other Medicines).

Venetoclax Toxicity: Concomitant administration of posaconazole with venetoclax (a CYP3A4 substrate) may increase venetoclax toxicities, including the risk of tumor lysis syndrome (TLS) and neutropenia (see Interactions with other medicines under Effects of Posaconazole on Other Drugs). Refer to the venetoclax prescribing information for detailed guidance.

Effects on Fertility:

Posaconazole had no effect on the fertility of male rats at doses up to 180 mg/kg/day (1.6

times the maximum recommended clinical dose (RCD) based on AUC at steady state in healthy volunteers fed a high fat meal). Like other azoles, male dogs administered oral posaconazole had findings consistent with reduced plasma testosterone levels, including spermatogenic giant cells (relative exposure 4.2). Posaconazole administered to female rats at doses up to 45 mg/kg/day (relative exposure 2.0) for 2 weeks prior to mating did not affect fertility, but disruption of oestrus cycling was seen in female rats treated for 4 weeks.

Use in Pregnancy:

Pregnancy Category B3 – There are no adequate studies in pregnant women. A total of three pregnancies have been reported in female subjects treated with posaconazole. Two pregnancies were electively terminated; no examination was reported on the fetuses. Another pregnancy was diagnosed at a follow-up visit approximately 1 month after the completion of a full 16-week prophylactic treatment with POS oral 200 mg TID in a patient who had received an allogeneic haematopoietic stem cell transplant. The subject delivered a healthy full-term male infant via caesarean section.

Studies in rats have shown reproductive toxicity including post implantation loss, increased skeletal variations, teratogenicity (craniofacial malformations), increased gestation length, dystocia, and reduced postnatal viability at exposure levels lower than those expected at the recommended doses in humans. An increase in post implantation loss and increased skeletal variations were seen in rabbits at plasma exposure levels greater than those of humans receiving therapeutic doses of posaconazole.

NOXAFIL must not be used during pregnancy unless the benefit to the mother clearly outweighs the risk to the foetus. Women of childbearing potential must be advised to always use effective contraceptive measure during treatment and for at least 2 weeks after completing therapy.

Use in Lactation:

Posaconazole is excreted in milk of lactating rats. The excretion of posaconazole in human breast milk has not been investigated. Women taking posaconazole should not breastfeed.

Use in the Elderly:

No dosage adjustment is recommended for geriatric patients (See Clinical Pharmacology – Pharmacokinetics in special populations – Elderly).

Effects on Adrenal Steroid Hormones:

As observed with other azole antifungal agents, effects related to inhibition of adrenal steroid hormone synthesis were seen in repeat-dose toxicity studies with posaconazole. Adrenal suppressive effects were observed in toxicity studies in rats and dogs at exposures equal to or greater than those obtained at therapeutic doses in humans.

Carcinogenicity:

Posaconazole caused an increase in hepatocellular adenomas in mice at plasma exposure levels ~7-times higher than anticipated in humans at the maximum recommended clinical dose. This finding is considered to have occurred secondary to liver toxicity in the species, and mice are known to be particularly susceptible to this neoplastic change.

Rats treated with posaconazole at exposure levels ≥ 2.4 -times that of humans developed adrenal cortical cell adenomas and/or carcinomas and phaeochromocytomas. The cortical tumours are consistent with endocrinological disruption following chronic impairment of adrenal steroidogenesis. The increase in phaeochromocytomas is considered to be a rat-specific phenomenon that follows changes in calcium homeostasis. Altered calcium homeostasis has not been observed in humans receiving posaconazole. The results of animal studies indicate little carcinogenic risk for posaconazole in clinical use.

Genotoxicity:

Posaconazole has been tested for genotoxicity in a series of *in vitro* assays (bacterial mutation, mammalian mutation and human lymphocyte chromosomal aberration) and an *in vivo* mouse micronucleus test. Under the conditions of these assays, posaconazole did not cause genetic damage.

Interactions with other medicines:

Summary of Drug Interactions

Contraindicated	Avoid concomitant use unless the benefit outweighs the risk	Dose adjustment of other medications and/or monitoring of adverse events	No dose adjustment required
Ergotamine or dihydroergotamine	Rifabutin	Rifabutin	Antacids

Terfenadine, astemizole	Phenytoin	Cyclosporine	Zidovudine, ritonavir, lamivudine, indinavir
Cisapride	Cimetidine	Tacrolimus	Glipizide
Pimozide	Efavirenz	Sirolimus	
Quinidine		Vinca alkaloids	
Halofantrine		Midazolam	
		Atazanavir/ritonavir	
		HMG-CoA reductase inhibitors metabolized through CYP3A4	
		Calcium channel blockers metabolized through CYP3A4	
		Sulfonylureas	
		Digoxin	

Note that the majority of the interaction studies were carried out in healthy volunteers with a repeat dose regimens of posaconazole 400 mg (suspension) twice daily administered with a meal or nutritional supplement. See below for further information.

Effect of Other Drugs on Posaconazole

Posaconazole is metabolized via UDP glucuronidation (phase 2 enzymes) and is a substrate for p-glycoprotein (P-gp) efflux. Therefore, inhibitors or inducers of these clearance pathways may affect posaconazole plasma concentrations.

Rifabutin (300 mg once a day) decreased the C_{max} (maximum plasma concentration) and AUC (area under the plasma concentration time curve) of posaconazole by 43 % and 49 %, respectively. Concomitant use of posaconazole and rifabutin should be avoided unless the benefit to the patient outweighs the risk.

Phenytoin (200 mg once a day) decreased the C_{max} and AUC of posaconazole by 41 % and 50 %, respectively. Concomitant use of posaconazole and phenytoin should be avoided unless the benefit to the patient outweighs the risk.

Cimetidine (400 mg twice a day) decreased the C_{max} and AUC of posaconazole 200 mg once a day each by 39 %. Concomitant use of posaconazole and cimetidine should be

avoided unless the benefit outweighs the risk. The effect of other H₂ receptor antagonists and proton pump inhibitors that may suppress gastric acidity has not been studied. Reduction in bioavailability may occur, therefore co- administration of posaconazole with H₂ receptor antagonists and proton pump inhibitors should be avoided if possible.

Antacids (20 mL single dose of liquid antacid equivalent to 25.4 mEq acid neutralizing capacity/5 mL) had no clinically significant effect on posaconazole C_{max} and AUC. No posaconazole dosage adjustments are required.

Glipizide: (10 mg single dose) had no clinically significant effect on posaconazole C_{max} and AUC. No posaconazole dosage adjustments are required.

Ritonavir (600 mg twice a day) had no clinically significant effect on posaconazole C_{max} and AUC. No posaconazole dosage adjustments are required.

Efavirenz: (400 mg once a day) decreased the C_{max} and AUC of posaconazole by 45 % and 50 %, respectively. Concomitant use of posaconazole and efavirenz should be avoided unless the benefit to the patient outweighs the risk.

Fosamprenavir: Combining fosamprenavir with posaconazole may lead to decreased posaconazole plasma concentrations. If concomitant administration is required, close monitoring for breakthrough fungal infections is recommended. Repeat dose administration of fosamprenavir (700 mg BID x 10 days) decreased the C_{max} and AUC of posaconazole (200 mg QD on the 1st day, 200 mg BID on the 2nd days, then 400 mg BID x 8 days) by 21 % and 23 % respectively.

Effects of Posaconazole on Other Drugs

Posaconazole is not metabolized to a clinically significant extent through the cytochrome P450 system. However, posaconazole is an inhibitor of CYP3A4 and thus the plasma levels of drugs that are metabolized through this enzyme pathway may increase when administered with posaconazole.

Terfenadine, astemizole, cisapride, pimozone, and quinidine: Although not studied *in vitro* or *in vivo*, co-administration of posaconazole and certain drugs such as terfenadine, astemizole, cisapride, pimozone, and quinidine, metabolized through the CYP3A4 system may result in increased plasma concentrations of these drugs, leading to potentially serious and/or life threatening adverse events (QT_c prolongation and rare occurrences of

torsade de pointes). Therefore, co- administration of these drugs with posaconazole is contraindicated (See Contraindications).

Ergot alkaloids: Although not studied *in vitro* or *in vivo*, posaconazole may increase the plasma concentration of ergot alkaloids (ergotamine and dihydroergotamine), which may lead to ergotism. Coadministration of posaconazole and ergot alkaloids is contraindicated (See Contraindications).

Vinca alkaloids: Most of the vinca alkaloids (e.g., vincristine and vinblastine) are substrates of CYP3A4. Concomitant administration of azole antifungals, including posaconazole, with vincristine has been associated with serious adverse reactions (see 5.2 Special Warnings and Special Precautions for Use). Posaconazole may increase the plasma concentrations of vinca alkaloids which may lead to neurotoxicity and other serious adverse reactions. Therefore, reserve azole antifungals, including posaconazole, for patients receiving a vinca alkaloid, including vincristine, who have no alternative antifungal treatment options.

Cyclosporine: In heart transplant patients on stable doses of cyclosporine, posaconazole 200 mg once daily increased cyclosporine concentrations requiring dose reductions. Cases of elevated cyclosporine levels resulting in serious adverse events, including nephrotoxicity and one fatal case of leukoencephalopathy, were reported in clinical efficacy studies. When initiating treatment with posaconazole in patients already receiving cyclosporine, the dose of cyclosporine should be reduced (e.g. to about three quarters of the current dose). Thereafter blood levels of cyclosporine should be monitored carefully during co-administration, and upon discontinuation of posaconazole treatment, and the dose of cyclosporine should be adjusted as necessary.

Tacrolimus: Posaconazole increased C_{max} and AUC of tacrolimus (0.05 mg/kg single dose) by 121 % and 358 %, respectively. Clinically significant interactions resulting in hospitalisation and/or posaconazole discontinuation were reported in clinical efficacy studies. When initiating posaconazole treatment in patients already receiving tacrolimus, the dose of tacrolimus should be reduced (e.g. to about one third of the current dose). Thereafter blood levels of tacrolimus should be monitored carefully during co-administration, and upon discontinuation of posaconazole, and the dose of tacrolimus should be adjusted as necessary.

Sirolimus: Repeat dose administration of oral posaconazole (400 mg twice daily for

16 days) increased the C_{max} and AUC of sirolimus (2 mg single dose) an average of 6.7-fold and 8.9 fold, respectively, in healthy subjects. When initiating therapy in patients already taking sirolimus, the dose of sirolimus should be reduced (e.g., to about 1/10 of the current dose) with frequent monitoring of sirolimus whole blood trough concentrations. Sirolimus concentrations should be performed upon initiation, during coadministration, and at discontinuation of posaconazole treatment, with sirolimus doses adjusted accordingly.

Rifabutin: Posaconazole increased the C_{max} and AUC of rifabutin by 31 % and 72 %, respectively. Concomitant use of posaconazole and rifabutin should be avoided unless the benefit to the patient outweighs the risk. If the drugs are coadministered, careful monitoring of full blood counts and adverse effects related to increased rifabutin levels (e.g., uveitis) is recommended.

Midazolam: Repeat dose administration of oral posaconazole 200 mg or 400 mg twice daily with a high fat meal, significantly increased the midazolam C_{max} by 2.2 fold (~7.03 to 15.4 ng/mL); AUC by approximately 5 fold (~31.9 to 159 h.ng/mL); and prolonged the mean terminal half-life of midazolam 8 to 10 hours in healthy subjects. It is recommended that dose adjustments of benzodiazepines, including midazolam metabolised by CYP3A4, be considered during co-administration with posaconazole.

Zidovudine (AZT), lamivudine (3TC), indinavir: In HIV infected patients on stable doses of zidovudine (300 mg twice a day or 200 mg every 8 hours), lamivudine (150 mg twice a day), and/or indinavir (800 mg every 8 hours), posaconazole had no clinically significant effect on the C_{max} and AUC of these medicinal products.

HMG-CoA reductase inhibitors primarily metabolized through CYP3A4: Repeat dose administration of oral posaconazole (50, 100 and 200 mg once daily for 13 days) increased the C_{max} and AUC of simvastatin (40mg single dose) an average of 7.4- to 11.4-fold, and 5.7- to 10.6-fold, respectively. Increased HMG-CoA reductase inhibitor concentrations in plasma can be associated with rhabdomyolysis. Coadministration of posaconazole and HMG-CoA reductase inhibitors primarily metabolized through CYP3A4 is contraindicated.

Calcium channel blockers metabolized through CYP3A4: Although not studied *in vitro* or *in vivo*, frequent monitoring for adverse effects and toxicity related to calcium channel blockers is recommended during coadministration with posaconazole. Dose adjustment of calcium channel blockers may be required.

Digoxin: Administration of other azoles has been associated with increases in digoxin levels. Therefore, posaconazole may increase plasma concentration of digoxin and digoxin levels need to be monitored when initiating or discontinuing posaconazole treatment.

Sulfonylureas: Glucose concentrations decreased in some healthy volunteers when glipizide was co-administered with posaconazole. Monitoring of glucose concentrations is recommended in diabetic patients.

HIV protease inhibitors: As HIV protease inhibitors are CYP3A4 substrates, it is expected that posaconazole will increase plasma levels of these antiretroviral agents. Repeat dose administration of oral posaconazole (400 mg twice daily for 7 days) increased the C_{max} and AUC of atazanavir (300 mg once a day for 7 days) an average of 2.6-fold and 3.7-fold, respectively, in healthy subjects. Repeat dose administration of oral posaconazole (400 mg twice daily for 7 days) increased the C_{max} and AUC of atazanavir to a lesser extent when administered as a boosted regimen with ritonavir (300 mg atazanavir plus ritonavir 100 mg once a day for 7 days) with an average of 1.5-fold and 2.5-fold, respectively, in healthy subjects. Frequent monitoring for adverse events and toxicity related to antiretroviral agents that are substrates of CYP3A4 is recommended during co-administration with posaconazole.

Venetoclax: Concomitant use of venetoclax (a CYP3A4 substrate) with posaconazole increases venetoclax C_{max} and AUC_{0-INF}, which may increase venetoclax toxicities (see PRECAUTIONS).

ADVERSE EFFECTS

The safety of posaconazole has been assessed in > 2400 patients and healthy volunteers enrolled in clinical trials and from post-marketing experience. The most frequently reported serious related adverse events included nausea, vomiting, diarrhea, pyrexia, and increased bilirubin.

Table 6. Treatment-related adverse events (TRAE) by body system and frequency <i>Common (≥1/100, <1/10); uncommon (≥1/1,000, <1/100); rare (≥1/10,000, <1/1,000)</i>	
Blood and lymphatic system disorders	
Common:	neutropenia

Uncommon:	thrombocytopenia, leukopenia, anaemia, eosinophilia, lymphadenopathy
Rare:	haemolytic uraemic syndrome, thrombotic thrombocytopenic purpura, pancytopenia, coagulopathy, haemorrhage
Immune system disorders	
Uncommon:	allergic reaction
Rare:	hypersensitivity reaction
Endocrine disorders	
Rare:	adrenal insufficiency, blood gonadotropin decreased
Metabolism and nutrition disorders	
Common:	electrolyte imbalance, anorexia
Uncommon:	hyperglycaemia
Psychiatric disorders	
Rare:	psychotic disorder, depression
Nervous system disorders	
Common:	paresthesia, dizziness, somnolence, headache
Uncommon:	convulsion, neuropathy, hypoaesthesia, tremor
Rare:	cerebrovascular accident, encephalopathy, peripheral neuropathy, syncope
Eye disorders	
Uncommon:	blurred vision
Rare:	diplopia, scotoma
Ear and labyrinth disorder	
Rare:	hearing impaired
Cardiac disorders	
Uncommon:	long QT syndrome ^s , electrocardiogram abnormal ^s , palpitations
Rare:	torsade de pointes, sudden death, ventricular tachycardia, cardio-respiratory arrest, cardiac failure, myocardial infarction
Vascular disorders	
Uncommon:	hypertension, hypotension
Rare:	pulmonary embolism, deep vein thrombosis
Respiratory, thoracic and mediastinal disorders	
Rare:	pulmonary hypertension, interstitial pneumonia, pneumonitis
Gastrointestinal disorders	
Common:	vomiting, nausea, abdominal pain, diarrhoea, dyspepsia, dry mouth, flatulence
Uncommon:	pancreatitis

Rare:	gastrointestinal haemorrhage, ileus
Hepatobiliary disorders	
Common:	liver function tests raised (ALT increased, AST increased, bilirubin increased, alkaline phosphatase increased, GGT increased)
Uncommon:	hepatocellular damage*, hepatitis, jaundice, hepatomegaly
Rare:	hepatic failure, hepatitis cholestatic, cholestasis, hepatosplenomegaly, liver tenderness, asterixis
Skin and subcutaneous tissue disorders	
Common:	rash
Uncommon:	mouth ulceration, alopecia
Rare:	Stevens Johnson syndrome, vesicular rash
Musculoskeletal and connective tissue disorders	
Uncommon:	back pain
Renal and urinary disorders	
Uncommon:	acute renal failure, renal failure, blood creatinine increased
Rare:	renal tubular acidosis, interstitial nephritis
Reproductive system and breast disorders	
Uncommon:	menstrual disorder
Rare:	breast pain
General disorders and administration site conditions	
Common:	pyrexia (fever), asthenia, fatigue
Uncommon:	oedema, pain, chills, malaise
Rare:	tongue oedema, face oedema
Investigations	
Uncommon:	medicine level changed

*During post-marketing surveillance severe hepatic injury with fatal outcome has been reported (see Precautions).
§ (see Precautions)

Post-marketing Experience

The following post-marketing adverse experience has been reported:

Endocrine Disorders: pseudoaldosteronism

DOSAGE AND ADMINISTRATION

Non-Interchangeability between Noxafil Modified Release Tablets and Noxafil Oral Suspension

Noxafil modified release tablets and oral suspension are not to be used interchangeably due to the differences in the dosing of each formulation. Therefore, follow the specific dosage recommendations for each of the formulations.

Important Administration Instructions for Noxafil Modified Release Tablet and Noxafil Oral Suspension

The prescriber should follow the specific dosing instructions for each formulation. The tablet and oral suspension are not to be used interchangeably due to the differences in the dosing of each formulation.

Shake well before use.

Refractory Invasive Fungal Infections (IFI) / Intolerant Patients with IFI: NOXAFIL should be administered at a dose of 400 mg (10 mL) twice a day with a meal or 240 mL of nutritional supplement. Dividing the dose further to 200 mg (5 mL) four times a day has been shown to enhance exposure to posaconazole, particularly in patients who have limited oral intake. Increasing the total daily dose above 800 mg does not further enhance the exposure to posaconazole (See Pharmacology).

Oropharyngeal Candidiasis in HIV-infected patients: NOXAFIL should be administered as a loading dose of 200 mg (5 mL) once a day on the first day, then 100 mg (2.5 mL) once a day for 13 days.

Oropharyngeal Candidiasis refractory to itraconazole or fluconazole in HIV- infected patients: NOXAFIL should be administered at a dose of 400 mg (10 mL) twice a day.

Prophylaxis of Invasive Fungal Infections: NOXAFIL should be administered at a dose of 200 mg (5 mL) three times a day. The duration of therapy is based on recovery from neutropenia or immunosuppression. For patients with acute myelogenous leukaemia or myelodysplastic syndromes, prophylaxis with Noxafil should start several days before the anticipated onset of neutropenia and continue for 7 days after the neutrophil count rises

above 500 cells/mm³.

Each dose of NOXAFIL should be administered with a meal, or with a nutritional supplement in patients who cannot tolerate food, to enhance exposure.

Duration of therapy should be based on the severity of the patient's underlying disease, recovery from immunosuppression, and clinical response.

There are limited pharmacokinetic data in patients with severe gastrointestinal dysfunction (such as severe diarrhoea). Patients who have severe diarrhoea or vomiting should be monitored closely for breakthrough fungal infections.

Use in renal impairment: No dose adjustment is required for renal dysfunction and as posaconazole is not significantly renally eliminated, an effect of severe renal insufficiency on the pharmacokinetics of posaconazole is not expected and no dose adjustment is recommended (See Pharmacology).

Use in hepatic impairment: There is limited pharmacokinetic data in patients with hepatic insufficiency; therefore, no recommendation for dose adjustment can be made. In the small number of subjects studied who had hepatic insufficiency, there was an increase in half-life with a decrease in hepatic function (See Pharmacology).

Use in the Elderly: No dosage adjustment is recommended for elderly patients (See Pharmacology).

Use in Children: Safety and efficacy in children and adolescents below the age of 18 years have not been established. Therefore, posaconazole is not recommended for use in patients below 18 years of age.

OVERDOSAGE

During clinical trials, patients who received posaconazole doses up to 1600 mg/day had no noted adverse reactions different from those reported with patients at the lower doses. In addition, accidental overdose was noted in one patient who took 1200 mg twice a day for 3 days. No adverse reactions were noted by the investigator.

In a trial of patients with severe haemodialysis-dependent renal dysfunction ($Cl_{cr} < 20$ mL/min), posaconazole was not removed by haemodialysis. Thus, haemodialysis is unlikely to be effective in removing posaconazole from the systemic circulation.

PRESENTATION AND STORAGE CONDITIONS

NOXAFIL oral suspension contains 40 mg posaconazole per mL of suspension.

NOXAFIL oral suspension 105 mL is packaged in a 123 mL amber Ph. Eur. Type IV glass bottle, closed with a plastic child-resistant closure. A measuring spoon, composed of clear polystyrene and graduated to measure 2.5 mL or 5 mL of the suspension, is provided with each bottle.

Shake well before use. Store below 30 °C (86 °F); excursions permitted to 15-30 °C (59-86 °F). Do not freeze.

Shelf life after first opening the container is 4 weeks.

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Packed by

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